

IMAT MOCK EXAM 11 (Legacy): Final Dress Rehearsal

60 questions · 100 minutes · scoring: +1.5 correct, -0.4 wrong, 0 blank · personal target: 42/90.0 · Mock exam · answer + worked solution follow every question

Each question is immediately followed by its answer and worked solution, so this paper is for review and paper study rather than blind simulation. Letters refer to the option order printed here; the online app shuffles options on every attempt.

Section A - Reading Skills, General Knowledge & Logical Reasoning (Q1-9)

1. Memories formed during the day are initially fragile. In deep sleep, patterns of activity recorded in the hippocampus replay the firing sequences that occurred while a task was being learned, and this replay coincides with brief rhythmic events that coordinate communication between the hippocampus and the cortex. In rodents, experimentally disrupting these replay events selectively impairs later performance on the learned task while leaving performance on other tasks intact. In humans, volunteers kept awake after memorising a list of words recall fewer items the next day than volunteers allowed to sleep, even when both groups are tested at the same interval after learning. On this evidence, sleep is not a passive interruption of mental life but an active part of how new memories become stable.

Which of the following statements is most strongly supported by the passage?

- A)** Staying awake after memorising a word list has no effect on next-day recall in humans.
- B)** People kept awake after learning remember more because less time has passed since learning.
- C)** Disrupting hippocampal replay impairs rodents' performance on every task that is tested.
- D)** All human memories are formed during deep sleep.
- E)** Sleep appears to play an active role in stabilising newly formed memories.

Answer: E

The passage itself concludes that sleep is not a passive interruption but an active part of how new memories become stable, and the replay, disruption and sleep deprivation findings all build toward exactly that claim. The claim that disrupting replay impairs performance on every task tested is contradicted by the word 'selectively' and by performance on other tasks remaining intact. The suggestion that awake volunteers remember more because less time has passed ignores the design detail that both groups are tested at the same interval after learning, and the sleepers in fact recall more.

2. Wind and solar generators share a property that conventional power plants do not: their output follows the weather rather than demand. On windy spring afternoons a grid may receive more electricity than it can use, while on still winter evenings, when demand peaks, renewable output can fall close to zero. Grid-scale batteries installed in recent years address part of this mismatch by shifting surplus electricity across a few hours, and they have become cheap enough to be deployed in large numbers. They cannot, however, shift a summer surplus to a winter deficit, because storing electricity for months remains far too expensive with current technology. Electricity systems that rely heavily on wind and solar therefore still need some combination of long-duration storage, flexible demand, deliberately overbuilt generation and backup plants that can be counted on whatever the weather.

Which of the following statements is most strongly supported by the passage?

- A) Flexible demand on its own can replace all backup generation in a renewable-based system.
- B) The output of wind and solar plants follows electricity demand more closely than that of conventional plants.
- C) Because batteries have become cheap, overbuilding renewable generation is no longer necessary.
- D) Batteries can help balance renewable supply and demand over periods of a few hours, but not over whole seasons.
- E) Grid-scale batteries are already able to store summer surplus electricity for use in winter.

Answer: D

The passage states that grid batteries shift surplus electricity across a few hours and have become cheap, but cannot move a summer surplus to a winter deficit because storing electricity for months remains far too expensive, which is precisely the claim that batteries help over hours but not over whole seasons. The opening sentence says renewable output follows the weather rather than demand, so the claim that it follows demand reverses the passage. Despite cheap batteries, the final sentence still lists long duration storage, flexible demand, overbuilt generation and backup plants as necessary, so none of those can be dropped.

3. (General knowledge) The Italian physician Camillo Golgi, who spent most of his career at the University of Pavia, shared the 1906 Nobel Prize in Physiology or Medicine, awarded for work on the structure of the nervous system. With which scientist did he share it?

- A) Ivan Pavlov
- B) Robert Koch
- C) Santiago Ramón y Cajal
- D) Emil von Behring
- E) Charles Sherrington

Answer: C

The decisive fact: the 1906 Nobel Prize in Physiology or Medicine was shared by Camillo Golgi and the Spanish histologist Santiago Ramón y Cajal for their work on the structure of the nervous system, Golgi for the silver staining 'black reaction' that made individual nerve cells visible and Cajal for using it to establish the neuron doctrine. Pavlov (1904), von Behring (1901) and Koch (1905) won in adjacent years for different work, and Sherrington's share came much later, in 1932.

4. (General knowledge) At the Milano Cortina 2026 Winter Olympic Games, held in Italy in February 2026, which country finished at the top of the final medal table?

- A) Norway B) United States C) Netherlands D) Germany E) Italy

Answer: A

The decisive fact: at the Milano Cortina 2026 Winter Games Norway finished at the top of the final medal table with a record 18 gold medals and 41 medals in total, extending its run of leading the Winter Games tables from 2018 and 2022. The United States (33 medals) and the Netherlands followed, and the host nation Italy, despite exceeding its own goals, did not lead the table.

5. Some buses in the city fleet are electric. All electric vehicles are exempt from the congestion charge. No vehicle that is exempt from the congestion charge is allowed to use the toll road. Which one of the following statements must be true?

- A) Some electric buses in the city fleet use the toll road.
- B) Some buses in the city fleet are not allowed to use the toll road.
- C) No bus in the city fleet is allowed to use the toll road.
- D) The buses in the city fleet that are not electric must pay the congestion charge.
- E) Every vehicle that is exempt from the congestion charge is electric.

Answer: B

Chain the premises: the electric buses in the fleet are electric vehicles, every electric vehicle is exempt from the congestion charge, and no exempt vehicle may use the toll road, so those electric buses certainly may not use the toll road. Since the fleet has some electric buses, some buses in the city fleet are therefore not allowed on the toll road. The claim that every exempt vehicle is electric is an illegitimate reversal of 'all electric vehicles are exempt', and the claim that some electric buses use the toll road contradicts the chain outright. Nothing constrains the non electric buses, so both the claim that they must pay the congestion charge and the stronger claim that no bus may use the toll road go beyond the evidence.

6. Five runners, Aisha, Bruno, Carmen, Devi and Lena, finished a 400 metre race, taking places 1 to 5 with no ties. Lena finished third. Carmen finished immediately behind Lena. Devi finished behind Carmen. Bruno did not finish first. Which one of the following statements must be true?

- A) Bruno finished ahead of Aisha.
- B) Bruno finished second.
- C) Aisha finished third.
- D) Devi finished fourth.
- E) Carmen finished immediately behind Devi.

Answer: B

Lena is third and Carmen finished immediately behind her, so Carmen is fourth. Devi finished behind Carmen, and the only place behind fourth is fifth, so Devi is fifth. That leaves first and second for Aisha and Bruno, and since Bruno did not finish first, Aisha won and Bruno was second. The order Aisha, Bruno, Lena, Carmen, Devi is the unique arrangement fitting every clue, so the only statement that must be true is that Bruno finished second; in this order Aisha is first rather than third, Devi is fifth rather than fourth, and Bruno finished behind Aisha, not ahead.

7. At a summer camp, each of the 100 children does at least one of three activities: swimming, sailing and archery. In total, 55 swim, 50 sail and 45 do archery; 25 swim and sail, 20 swim and do archery, 15 sail and do archery; and 10 do all three. How many children do exactly two of the three activities?

- A) 20 B) 25 C) 30 D) 35 E) 40

Answer: C

Step 1: subtract the triple overlap from each pairwise figure to get the 'exactly two' counts: swim and sail only = $25 - 10 = 15$, swim and archery only = $20 - 10 = 10$, sail and archery only = $15 - 10 = 5$.

Step 2: exactly two activities = $15 + 10 + 5 = 30$

Check: the only swimming, only sailing and only archery regions are 20, 20 and 20, and $20 + 20 + 20 + 15 + 10 + 5 + 10 = 100$, matching the 100 children. The value 40 counts children doing at least two activities, because it keeps the ten who do all three inside the tally; exactly two requires removing them from every pair.

8. 1 April 2026 was a Wednesday. What was the date of the second Monday of April 2026?

- A) 5 April B) 6 April C) 12 April D) 13 April E) 20 April

Answer: D

Step 1: 1 April 2026 was a Wednesday, so counting forward gives 2 April Thursday, 3 Friday, 4 Saturday, 5 Sunday, 6 April Monday: the first Monday is 6 April. Step 2: Mondays recur every seven days, so the Mondays of April 2026 are 6, 13, 20 and 27 April. Step 3: the second Monday is therefore 13 April. The date 6 April is only the first Monday, and 12 April is a Sunday, one day short.

9. A cube is folded from a flat net of six identical numbered squares. The net consists of one column of four squares numbered 5, 1, 3 and 6 from top to bottom, with the square numbered 2 attached to the left edge of square 1 and the square numbered 4 attached to the right edge of square 1. When the net is folded into a cube, which number is on the face opposite the face numbered 1?

- A) 2 B) 3 C) 4 D) 5 E) 6

Answer: E

The four squares in the column (5, 1, 3, 6 from top to bottom) fold into a band that wraps around four faces of the cube, and in any such band faces two positions apart end up opposite, so 5 opposes 3 and 1 opposes 6. The two squares attached at the sides of face 1 fold over and end up opposite each other. Hence the face opposite the face numbered 1 carries the number 6. Every other candidate number (2, 3, 4, 5) is the label of a face adjacent to face 1 in the folded cube.

Section B - Biology (Q10-32)

10. Which statement about the enzymes and DNA synthesis at a bacterial replication fork is correct?

- A) The leading strand is synthesised continuously from a single primer, while the lagging strand is synthesised in short fragments as the fork opens.
B) The leading strand requires a new primer for each fragment it synthesises.
C) DNA ligase unwinds the parental double helix ahead of the replication fork.
D) Helicase synthesises the short RNA primers needed to start DNA synthesis.
E) DNA polymerase can begin synthesis of a new strand without needing a primer.

Answer: A

Step 1: helicase unwinds the parental double helix, primase lays the short RNA primers, DNA polymerase can only extend an existing primer (it cannot start a chain), and ligase seals the gaps between adjacent fragments. Step 2: because the two template strands are antiparallel, only one new strand can be made continuously in the direction of fork opening, from a single primer; the other is built in short fragments, each started by its own primer. The claim that helicase makes the primers gives that job to the wrong enzyme (it belongs to primase), and the claim that the leading strand needs a new primer for every fragment describes the lagging strand instead.

11. Which statement about transcription in a eukaryotic cell is correct?

- A)** Transcription begins when RNA polymerase binds to the terminator sequence of a gene.
- B)** RNA polymerase synthesises the new RNA strand in the 3' to 5' direction.
- C)** The poly-A tail is a stretch of DNA sequence copied from the end of the gene.
- D)** RNA polymerase binds to the promoter and starts RNA synthesis without a primer, using one DNA strand as a template.
- E)** RNA polymerase requires an RNA primer before it can start copying a gene.

Answer: D

RNA polymerase binds the promoter sequence (the terminator instead signals where transcription stops) and initiates RNA synthesis without any primer, unlike DNA polymerase, while reading one DNA strand as the template. The new RNA grows in the 5' to 3' direction, so the claim that synthesis runs 3' to 5' reverses the actual direction. The poly-A tail is not transcribed from the gene; it is added enzymatically to the 3' end of the finished pre-mRNA in the nucleus.

12. Which statement about the ribosome during translation is correct?

- A)** In eukaryotes, ribosomes begin translating an mRNA while that mRNA is still being transcribed.
- B)** A tRNA carrying the next amino acid enters at the A site, where its anticodon pairs with the mRNA codon.
- C)** The initiator tRNA binds first to the A site of the ribosome when translation starts.
- D)** The E site is where peptide bonds are formed between amino acids.
- E)** The growing polypeptide chain is carried on a tRNA sitting in the A site while new tRNAs enter at the P site.

Answer: B

During elongation a tRNA carrying the next amino acid enters at the A (aminoacyl) site of the ribosome and its anticodon base pairs with the mRNA codon exposed there; peptidyl transferase then transfers the growing chain onto it, and tRNAs step through the P (peptidyl) site and leave by the E (exit) site. The initiator tRNA starts in the P site, not the A site, peptide bonds form at the peptidyl transferase centre of the large subunit rather than at the exit site, and the growing chain rides on the P site tRNA, not on newly entering ones. Bacteria can translate an mRNA while it is still being transcribed, but eukaryotes cannot, since their transcription happens in the nucleus and translation in the cytoplasm.

13. The beginning of an mRNA reads 5' AUG GCA UAC GGU UAA 3', where AUG is the start codon and UAA is a stop codon. Codon meanings: AUG = methionine, GCA = alanine, GCC = alanine, GUA = valine, UAC = tyrosine, GGU = glycine, CGG = arginine, UUA = leucine. Two independent mutations occur in different copies of this gene. Mutation 1 substitutes one base so that codon 2 reads GCC instead of GCA. Mutation 2 inserts a single G immediately after codon 2. Which statement about the resulting proteins is correct?

- A)** Mutation 2 adds exactly one extra amino acid to the protein and leaves all the other amino acids unchanged.
- B)** Both mutations are silent, because neither changes any amino acid in the protein.
- C)** Mutation 1 is silent, while mutation 2 shifts the reading frame, so from codon 3 onward the mRNA reads GUA, CGG and UUA instead of UAC, GGU and UAA, changing the amino acids and losing the stop codon.
- D)** Mutation 1 changes the amino acid at position 2 of the protein, while mutation 2 is silent.
- E)** Mutation 1 is silent and mutation 2 changes only the amino acid encoded by codon 2.

Answer: C

Step 1: mutation 1 changes codon 2 from GCA to GCC, and both codons specify alanine, so it is silent. Step 2: mutation 2 inserts one base after codon 2, shifting the reading frame: the tail UAC GGU UAA now regroups as GUA, CGG, UUA, which encode valine, arginine and leucine in place of tyrosine, glycine and the stop signal, so every amino acid from position 3 changes and translation runs past the old stop. The suggestion that an insertion adds exactly one amino acid or changes only one position confuses an insertion with a substitution; only inserting or deleting a multiple of three bases would leave the downstream reading frame intact.

14. In a testcross, a pea plant heterozygous for three independently assorting genes, AaBbCc, is crossed with a plant that is aabbcc. What is the probability that an offspring is heterozygous at all three loci (AaBbCc)?

- A)** 1/16 **B)** 1/8 **C)** 1/4 **D)** 1/2 **E)** 3/4

Answer: B

The testcross parent aabbcc can contribute only recessive alleles, so at each locus the cross is effectively Aa × aa, giving heterozygous offspring with probability 1/2 and homozygous recessive offspring with probability 1/2. The three genes assort independently, so the probabilities multiply: $1/2 \times 1/2 \times 1/2 = 1/8$. The ratio 3/4 belongs to a heterozygote self cross, where three of four offspring show the dominant phenotype, and has no place in a testcross.

15. Cystic fibrosis is caused by an autosomal recessive allele. Two parents without the condition have one child with cystic fibrosis and a second child who is healthy. What is the probability that this healthy second child is a carrier of the allele?

- A)** 1/4 **B)** 1/3 **C)** 1/2 **D)** 2/3 **E)** 3/4

Answer: D

An affected child from two unaffected parents proves that both parents are carriers, so each pregnancy gives 1/4 AA, 1/2 Aa and 1/4 aa. Conditioning on the child being healthy removes only the aa class, leaving one AA to two Aa among unaffected offspring, so the probability that the healthy child is a carrier is 2/3. The unconditional carrier risk for any child of carrier parents is 1/2, which ignores the given information that this particular child is unaffected.

16. Haemophilia is caused by a recessive allele on the X chromosome. A woman has no symptoms, but her father had haemophilia. She and her husband, who does not have haemophilia, are expecting their first child. What is the probability that the first child is a daughter who carries the allele?

- A)** 1/8 **B)** 1/4 **C)** 1/2 **D)** 1 **E)** 3/4

Answer: B

The woman's father passed his only X chromosome, which carried the allele, to all his daughters, so she is certainly a carrier; her unaffected husband passes a normal X to every child. The probability that the first child is a daughter is 1/2, and a daughter of a carrier mother has a 1/2 chance of receiving the mother's mutant X, giving $1/2 \times 1/2 = 1/4$. Stopping at 1/2 forgets to include the chance that the child is a girl at all.

17. In a series of crosses, the recombination frequencies between three linked genes are: between gene a and gene b, 4 percent; between gene b and gene c, 9 percent; between gene a and gene c, 13 percent. Which conclusion is correct?

- A)** The genes are arranged with gene b between the other two, in the order a b c.
B) The genes are arranged with gene a between the other two, in the order b a c.
C) No valid order exists, because the recombination frequency between two genes must be the same whichever third gene is considered.
D) The genes are arranged with gene c between the other two, in the order a c b.
E) The three genes must lie on three different chromosomes.

Answer: A

Since $13 = 4 + 9$, the largest recombination frequency is the sum of the other two, which is the signature of the middle gene lying between the outer two: the map order is a, then b 4 units away, then c a further 9 units away. If a or c were the middle gene, the largest distance would have to equal the difference of the other two, not the sum. Genes on three different chromosomes would assort independently and give pairwise recombination frequencies near 50 percent, not 4, 9 and 13 percent.

18. Which event occurs during anaphase of mitosis in an animal cell?

- A)** Nuclear envelopes re-form around each group of chromosomes.
B) Chromosomes align along the equator of the cell.
C) The nuclear envelope breaks down and the spindle begins to form.
D) Homologous chromosome pairs separate and move to opposite poles.
E) Sister chromatids separate at the centromeres and move toward opposite poles.

Answer: E

Anaphase of mitosis begins when the sister chromatids separate at their centromeres and are pulled toward opposite spindle poles. The separation of homologous chromosome pairs happens in anaphase I of meiosis, not mitosis; alignment at the cell's equator is metaphase; breakdown of the nuclear envelope belongs to prophase; and re-formation of nuclear envelopes around each chromosome group is telophase.

19. During meiosis II in a human oocyte, the sister chromatids of chromosome 21 fail to separate, while meiosis I and the rest of meiosis II proceed normally. One of the resulting egg cells receives both copies of chromosome 21 and is then fertilised by a normal sperm. Which statement is correct?

- A)** The zygote has 45 chromosomes, with only one copy of chromosome 21.
- B)** The zygote has 69 chromosomes because all chromosomes are present in triplicate.
- C)** The zygote has 47 chromosomes, with three copies of chromosome 21, the condition known as Down syndrome.
- D)** The zygote has 46 chromosomes and is chromosomally normal.
- E)** The zygote has 47 chromosomes, but the extra chromosome is a sex chromosome.

Answer: C

Step 1: nondisjunction of sister chromatids in meiosis II sends both copies of chromosome 21 into one egg (24 chromosomes) and none into the other, while meiosis I has already halved the set normally. Step 2: fertilisation of the egg carrying both copies by a normal sperm (23 chromosomes) gives $24 + 23 = 47$ chromosomes, with three copies of chromosome 21, which is trisomy 21, Down syndrome. All other chromosomes are present in the usual two copies, so totals of 45, 46 or 69 are impossible, and the extra chromosome is chromosome 21, not a sex chromosome.

20. Before industrialisation, light coloured peppered moths rested on pale, lichen covered tree trunks and greatly outnumbered dark coloured moths. As industrial soot blackened the trunks, the dark form became far more common within a few decades. Which statement best explains this change?

- A)** The dark form arose by random mutation before the trunks darkened; once the trunks became dark, birds caught more light moths, so dark moths survived and reproduced more successfully and passed on their alleles.
- B)** Soot chemically altered the genes of the moths so that their offspring were born dark.
- C)** Individual moths gradually darkened during their own lifetimes as soot settled on the trees.
- D)** The moths sensed the darkening trunks and produced dark offspring in order to blend into the background.
- E)** Dark coloured moths were already more common than light moths before the trees darkened.

Answer: A

The dark form arises by random mutation before the environment changes; once soot darkened the trunks, birds caught the more visible light moths, so dark moths survived and reproduced more successfully and passed on their alleles, which is natural selection acting on pre-existing variation. The alternative routes fail: individual moths cannot acquire heritable darkening during their own lifetimes, soot does not chemically rewrite genes, moths cannot produce dark offspring 'in order to' match a background, and the passage states that light moths greatly outnumbered dark ones before the trees darkened.

21. A population of diploid flowering plants grows in one meadow with no physical barriers. A mutation arises in one lineage that doubles the chromosome number, producing tetraploid plants. Tetraploids mate successfully with other tetraploids, but crosses between tetraploid and diploid plants produce seeds that die before germination. Which statement is correct?

- A)** A change in chromosome number can alter gene expression but can never give rise to a new species.
- B)** This is allopatric speciation, because the tetraploid and diploid plants cannot interbreed.
- C)** The tetraploid and diploid plants still belong to one biological species, because they live in the same meadow.
- D)** No speciation can occur without a physical barrier such as a river or a mountain range.
- E)** This is sympatric speciation: a reproductively isolated polyploid lineage has formed within the same geographic area.

Answer: E

The tetraploid plants arose within the same meadow as the diploids, so there is no geographic barrier, which rules out any allopatric account and the claim that a physical barrier is always required. Because tetraploid × diploid crosses produce seeds that die, the polyploids are reproductively isolated, so a new lineage has formed in sympatry, and polyploidy is the classic one step route to speciation in plants. Sharing a meadow does not by itself make one biological species: the species criterion is interbreeding, not geography, and a change in chromosome number clearly can create a new species here.

22. Which statement about the role of water in the light dependent reactions of photosynthesis is correct?

- A)** Electrons released by the splitting of water are passed directly to photosystem I to replace its lost electrons.
- B)** The oxygen released by photosynthesis comes from carbon dioxide that is split in the thylakoid membranes.
- C)** The oxygen released by photosynthesis comes from water molecules split during the light dependent reactions.
- D)** Water is split in the stroma as part of the Calvin cycle.
- E)** The splitting of water directly produces the ATP that the Calvin cycle consumes.

Answer: C

The oxygen released in photosynthesis comes from the splitting (photolysis) of water at photosystem II in the thylakoid membranes: the water's electrons replace those lost by the photosystem and its oxygen atoms are released as O₂. Electrons from water pass to photosystem II first, not directly to photosystem I; the splitting happens in the thylakoid membrane, not in the stroma, where the Calvin cycle instead fixes CO₂; and ATP is made afterwards by chemiosmosis, not directly by the splitting step. The claim that the released oxygen comes from carbon dioxide has the source backwards.

23. Which statement about long distance transport in flowering plants is correct?

- A)** Xylem transports water and dissolved mineral ions upward from the roots, while phloem transports sugars from source tissues to sink tissues.
- B)** The contents of both xylem and phloem are moved mainly by the transpiration pull from the leaves.
- C)** Phloem can transport sugars only downward, from the leaves to the roots.
- D)** Mature xylem vessels consist of living cells that actively pump water from cell to cell.
- E)** Xylem transports sugars from the leaves to the roots, while phloem transports water and mineral ions upward.

Answer: A

Xylem, made of dead hollow vessels, carries water and dissolved mineral ions upward from the roots, driven mainly by transpiration pull, while phloem's living sieve tubes carry sucrose from source tissues such as mature leaves to sink tissues such as roots, fruits and buds, and the phloem flow can run upward or downward. Swapping the two cargoes inverts the roles of the tissues, transpiration pull does not move phloem contents, and mature xylem is dead rather than living and actively pumping. The claim that sugars move only downward fails because sinks such as growing shoots sit above the leaves.

24. In a healthy heart, the ventricles begin to contract shortly after the atria contract. Which statement explains this timing?

- A)** The atria and the ventricles are excited at the same moment, but ventricular muscle contracts more slowly than atrial muscle.
- B)** The ventricles contract first and the atria contract during the diastole that follows.
- C)** A second impulse from the sinoatrial node travels directly to the ventricles and triggers their contraction.
- D)** The delay is caused by the time blood takes to flow from the atria into the ventricles.
- E)** The impulse from the sinoatrial node is delayed at the atrioventricular node, so the ventricles are excited shortly after the atria, allowing the atria to finish emptying before the ventricles contract.

Answer: E

Step 1: the impulse from the sinoatrial node spreads across the atria, which contract first. Step 2: conduction is delayed briefly at the atrioventricular node before the impulse continues through the bundle of His and the Purkinje fibres to the ventricles. Step 3: this electrical delay, not the time blood takes to flow and not a difference in muscle contraction speed, lets the atria finish emptying before ventricular contraction begins. There is only one impulse initiating each beat, and the ventricles never contract first in a healthy heart.

25. Which statement about ultrafiltration at the glomerulus is correct?

- A)** Plasma proteins pass freely into the filtrate and are reabsorbed later along the loop of Henle.
- B)** Red blood cells normally enter the filtrate and are returned to the blood in the distal tubule.
- C)** Glucose is actively transported from the blood into the filtrate at the glomerulus.
- D)** Fluid is forced from the glomerular capillaries into Bowman's capsule by blood pressure, and the filtrate normally contains glucose, urea and ions but no plasma proteins or blood cells.
- E)** Water is moved out of the glomerular capillaries by active transport.

Answer: D

Hydrostatic blood pressure in the glomerular capillaries forces water and small solutes (glucose, urea, ions, amino acids) through the filtration barrier into Bowman's capsule, while plasma proteins and blood cells are too large to cross and remain in the blood. Glucose enters the filtrate by filtration and is then reabsorbed in the proximal convoluted tubule; it is not actively secreted into the filtrate, and water leaves the capillaries under pressure rather than by active transport. The idea that proteins are filtered freely and reclaimed later describes a leakiness a healthy glomerulus does not have, and blood cells never normally reach the filtrate.

26. Which statement about nerve action potentials is correct?

- A)** Subthreshold stimuli produce small action potentials that travel more slowly along the axon.
- B)** Once threshold is reached, the amplitude of the action potential is fixed; a stronger stimulus increases the frequency of action potentials, not their size.
- C)** A stronger stimulus produces a single action potential with a larger amplitude.
- D)** During repolarisation, the inside of the axon becomes even more positive than at the peak of depolarisation.
- E)** The refractory period allows action potentials to travel backwards along the axon as well as forwards.

Answer: B

Action potentials are all or none events: once threshold is reached, every impulse has the same fixed amplitude, and stimulus strength is encoded as firing frequency, so a stronger stimulus produces more impulses per second, never larger ones. Subthreshold stimuli produce no action potential at all, rather than small slow ones. The refractory period guarantees one way propagation by preventing immediate re-excitation behind the impulse, and repolarisation carries the membrane potential back down toward the negative resting value, not above the depolarisation peak.

27. Which statement about the control of cortisol secretion is correct?

- A)** Cortisol raises its own secretion through a positive feedback loop.
- B)** When cortisol levels fall, the anterior pituitary releases less ACTH so that cortisol stays low.
- C)** ACTH travels down nerve fibres from the hypothalamus to the adrenal cortex.
- D)** A rise in blood cortisol acts by negative feedback on the hypothalamus and anterior pituitary, reducing the release of CRH and ACTH.
- E)** Cortisol is released by the adrenal medulla in response to nerve signals from the hypothalamus.

Answer: D

The hypothalamus secretes CRH, the anterior pituitary responds by releasing ACTH into the bloodstream, and ACTH stimulates the adrenal cortex to secrete cortisol; rising cortisol then inhibits CRH and ACTH release, a negative feedback loop. Cortisol comes from the cortex, not the medulla, which secretes adrenaline under direct nerve control; a fall in cortisol would release the brake and increase, not decrease, ACTH output; and ACTH is a blood borne hormone, not a signal travelling down nerve fibres. The loop is negative, not positive, feedback.

28. Which statement about B and T lymphocytes is correct?

- A)** Both B and T lymphocytes complete their maturation in the thymus.
- B)** T lymphocytes are the main antibody secreting cells of the immune system.
- C)** Activated B lymphocytes differentiate into plasma cells that secrete antibodies, whereas cytotoxic T cells destroy infected body cells by direct contact.
- D)** Cytotoxic T cells engulf bacteria by phagocytosis and digest them internally.
- E)** B lymphocytes mature in the thymus before circulating in the blood.

Answer: C

B lymphocytes mature in the bone marrow and, when activated, differentiate into plasma cells that secrete antibodies; T lymphocytes mature in the thymus, and cytotoxic T cells kill infected host cells by direct contact. Antibody secretion is the job of B cells, not T cells; phagocytosis of bacteria belongs to macrophages and neutrophils, not cytotoxic T cells; and only the T cells complete their maturation in the thymus, so placing B cell maturation there misassigns the site.

29. Which statement about the cytoskeleton is correct?

- A)** Microfilaments are hollow tubes of tubulin that give cilia their characteristic internal arrangement.
- B)** Microtubules are permanently fixed structures that never assemble or disassemble inside the cell.
- C)** Actin microfilaments form the spindle fibres that separate chromosomes during mitosis.
- D)** The contractile ring that divides an animal cell during cytokinesis is made of microtubules.
- E)** Microtubules are hollow polymers of tubulin that form the mitotic spindle and the core of cilia and flagella.

Answer: E

Microtubules are hollow polymers of the protein tubulin that build the mitotic spindle and form the axoneme at the core of cilia and flagella. The claim that the contractile ring dividing an animal cell at cytokinesis is made of microtubules, and the claim that actin microfilaments form the spindle fibres, each swap the real roles of microtubules and actin: the ring is actin, the spindle is tubulin. Calling microtubules permanently fixed structures also contradicts their constant dynamic assembly and disassembly.

30. Which statement about apoptosis is correct?

- A)** Apoptosis is a genetically regulated process in which the cell is dismantled into fragments that nearby cells engulf without causing inflammation.
- B)** Cells dying by apoptosis release their digestive enzymes into the surrounding tissue.
- C)** Apoptosis plays no role in healthy embryonic development.
- D)** Apoptosis is triggered only by external toxins and never by signals arising within the cell itself.
- E)** Apoptosis is the uncontrolled death of cells after injury, and it always triggers inflammation.

Answer: A

Apoptosis is programmed cell death: internal or external signals activate regulated pathways that fragment the DNA, shrink the cell and package it into membrane bound fragments that neighbouring cells engulf, so no digestive enzymes spill into the tissue and inflammation is avoided. It is essential in healthy development, for instance in removing the webbing between developing fingers. Uncontrolled death after injury with enzyme release and inflammation describes necrosis, and apoptosis is commonly triggered by signals arising within the cell, not only by external toxins.

31. Which statement correctly matches a level of protein structure with what defines it?

- A)** The primary structure is defined by hydrogen bonding between atoms of the polypeptide backbone.
- B)** The tertiary structure is the assembly of several separate polypeptide chains into one functional unit.
- C)** The secondary structure is defined by disulfide bridges between cysteine residues.
- D)** The primary structure is the sequence of amino acids joined together by peptide bonds.
- E)** The quaternary structure is the local coiling of the chain into alpha helices.

Answer: D

Primary structure is the linear sequence of amino acids joined by covalent peptide bonds. Hydrogen bonding between backbone atoms defines secondary structure (alpha helices and beta sheets), disulfide bridges between cysteines stabilise the tertiary fold rather than the secondary structure, and the assembly of several separate polypeptide chains into one functional unit is quaternary, not tertiary, structure. Calling the local coiling into alpha helices quaternary structure assigns a secondary feature to the wrong level.

32. A person standing outside in cold weather starts to shiver. Which statement about this response is correct?

- A)** Shivering is involuntary contraction of skeletal muscles that generates heat, coordinated by the hypothalamus as part of a negative feedback response to falling body temperature.
- B)** Skin blood vessels widen so that warm blood flows close to the surface, warming the skin.
- C)** Sweat production increases so that evaporation can release heat from the body core.
- D)** Shivering is a voluntary response that the person chooses to start in order to warm up.
- E)** Shivering cools the body further, because the contracting muscles push warm blood to the skin surface where heat is lost.

Answer: A

Falling body temperature is detected and the hypothalamus, the thermoregulatory centre, mounts heat generating and heat conserving responses: shivering, the involuntary rapid contraction of skeletal muscle whose metabolism releases heat, together with vasoconstriction of skin vessels and reduced sweating, all part of a negative feedback loop restoring core temperature. Shivering is not a voluntary choice, skin vessels constrict rather than widen in the cold, and increased sweating is the cooling response to heat, not to cold.

Section C - Chemistry (Q33-47)

33. What is the mass of 0.25 mol of copper(II) sulfate, CuSO_4 ? (Cu = 64, S = 32, O = 16)

- A)** 32 g **B)** 40 g **C)** 64 g **D)** 160 g **E)** 640 g

Answer: B

$$M(\text{CuSO}_4) = 64 (\text{Cu}) + 32 (\text{S}) + 4 \times 16 (\text{O}) = 160 \text{ g/mol}$$

$$\text{Mass} = n \times M = 0.25 \text{ mol} \times 160 \text{ g/mol} = 40 \text{ g}$$

The figure 160 g is the mass of a whole mole, and 640 g comes from dividing 160 by 0.25 instead of multiplying.

34. 28 g of nitrogen gas (N_2 , $M = 28 \text{ g/mol}$) is mixed with 9.0 g of hydrogen gas (H_2 , $M = 2.0 \text{ g/mol}$) and the mixture is reacted according to: $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightarrow 2\text{NH}_3(\text{g})$ (NH_3 , $M = 17 \text{ g/mol}$). What is the maximum mass of ammonia that can be produced?

- A)** 2.0 g **B)** 17 g **C)** 34 g **D)** 51 g **E)** 68 g

Answer: C

Step 1: moles are $28/28 = 1.0 \text{ mol}$ of N_2 and $9.0/2.0 = 4.5 \text{ mol}$ of H_2

Step 2: the equation needs 3 mol of H_2 per mol of N_2 , so 1.0 mol of N_2 requires only 3.0 mol of H_2 ; with 4.5 mol available, hydrogen is in excess and N_2 is the limiting reagent.

Step 3: 1.0 mol of N_2 yields 2.0 mol of NH_3 , so the maximum mass is $2.0 \times 17 = 34 \text{ g}$

The value 51 g follows from wrongly treating hydrogen as the limiting reagent and converting all 4.5 mol of it.

35. An oxide of sulfur is found by analysis to contain 50.0% sulfur and 50.0% oxygen by mass. What is its empirical formula? (S = 32, O = 16)

- A)** S_2O **B)** SO **C)** SO_4 **D)** SO_3 **E)** SO_2

Answer: E

Step 1: take 100 g of oxide: 50.0 g of S is $50.0/32 = 1.56 \text{ mol}$ and 50.0 g of O is $50.0/16 = 3.13 \text{ mol}$

Step 2: divide both by the smaller amount: $1.56 : 3.13 \approx 1 : 2$. The empirical formula is SO_2 . A 1:1 mole ratio would require equal numbers of moles of the two elements, but equal masses do not give that here, since an oxygen atom is only half the mass of a sulfur atom.

36. 5.85 g of sodium chloride ($M = 58.5 \text{ g/mol}$) is dissolved in water and the solution is made up to exactly 500 cm^3 . What is the concentration of the solution?

- A)** 0.050 mol/dm^3 **B)** 0.10 mol/dm^3 **C)** 0.20 mol/dm^3 **D)** 0.40 mol/dm^3 **E)** 0.50 mol/dm^3

Answer: C

Step 1: $n = m/M = 5.85/58.5 = 0.10 \text{ mol of NaCl}$

Step 2: $V = 500 \text{ cm}^3 = 0.500 \text{ dm}^3$

Step 3: $c = n/V = 0.10/0.500 = 0.20 \text{ mol/dm}^3$

The value 0.10 mol/dm^3 is just the mole number with the units swapped, ignoring the dilution to half a cubic decimetre.

37. Potassium hydroxide is a strong base that dissociates completely in water. What is the pH at 25°C of a $1.0 \times 10^{-4} \text{ mol/dm}^3$ solution of KOH? ($K_w = 1.0 \times 10^{-14}$)

- A)** 4 **B)** 7 **C)** 10 **D)** 11 **E)** 14

Answer: C

KOH is a strong base and dissociates completely, so $[\text{OH}^-] = 1.0 \times 10^{-4} \text{ mol/dm}^3$ and $\text{pOH} = 4$.

At 25°C , $\text{pH} + \text{pOH} = 14$, so $\text{pH} = 14 - 4 = 10$

The value 4 is the pOH mislabelled as pH, and a solution of a strong base can be neither acidic (pH 4) nor exactly neutral (pH 7).

38. Hydrochloric acid is titrated with sodium hydroxide solution. Which of the following statements is correct?

- A)** At the equivalence point neither reactant is in excess; since the salt NaCl does not hydrolyse, the solution has pH 7 at 25°C .
B) The equivalence point is defined as the point at which the indicator changes colour.
C) At the equivalence point the acid is still in excess, so the pH is below 7.
D) At the equivalence point, equal masses of acid and base have been added.
E) At the equivalence point of any titration, weak or strong, the pH is exactly 7.

Answer: A

At the equivalence point the moles of H^+ and OH^- have reacted exactly, and since NaCl is the salt of a strong acid and a strong base it does not hydrolyse, leaving the solution neutral at pH 7 at 25°C .

Equivalence means equal moles, not equal masses, of acid and base, and the indicator's colour change defines the endpoint, a related but distinct point. The claim that every equivalence point sits at pH 7 fails for weak acid or weak base systems: titrating a weak acid with NaOH gives an alkaline equivalence point because the conjugate base hydrolyses.

39. In acidic solution, permanganate ions oxidise ethanedioate (oxalate) ions: $\text{MnO}_4^- + \text{C}_2\text{O}_4^{2-} + \text{H}^+ \rightarrow \text{Mn}^{2+} + \text{CO}_2 + \text{H}_2\text{O}$. When the equation is balanced with the smallest whole-number coefficients, what is the coefficient of H^+ ?

- A)** 2 **B)** 5 **C)** 10 **D)** 16 **E)** 32

Answer: D

Step 1: half equations: $\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^- \rightarrow \text{Mn}^{2+} + 4\text{H}_2\text{O}$ (manganese falls from +7 to +2, gaining 5 electrons) and $\text{C}_2\text{O}_4^{2-} \rightarrow 2\text{CO}_2 + 2\text{e}^-$ (each carbon rises from +3 to +4, so 2 electrons lost per ion). Step 2: equalise the electrons at 10 by multiplying the first half equation by 2 and the second by 5. Step 3: adding gives $2\text{MnO}_4^- + 5\text{C}_2\text{O}_4^{2-} + 16\text{H}^+ \rightarrow 2\text{Mn}^{2+} + 10\text{CO}_2 + 8\text{H}_2\text{O}$; the charges balance (left: $-2 - 10 + 16 = +4$; right: $+4$), so the coefficient of H^+ is 16. The value 8 belongs to the unmultiplied permanganate half equation and ignores the factor of 2 needed to match the 10 electrons from five oxalate ions.

40. Which of the following statements about the octet rule is correct?

- A) BF_3 violates the octet rule because boron carries more than eight valence electrons.
- B) Boron in BF_3 has only six electrons in its valence shell, so BF_3 can accept an electron pair and act as a Lewis acid.
- C) Phosphorus in PCl_5 obeys the octet rule, because each chlorine atom shares one electron with it.
- D) Compounds of period 3 elements such as PCl_5 and SF_6 cannot exist because they exceed the octet.
- E) Nitrogen in NH_3 has an expanded octet of ten electrons.

Answer: B

Boron has three valence electrons and forms three bonds in BF_3 , leaving only six electrons in its valence shell; the molecule is electron deficient and readily accepts a lone pair, which is exactly the behaviour of a Lewis acid. PCl_5 (10 electrons around phosphorus) and SF_6 (12 around sulfur) do exist, because period 3 atoms can expand their octets, so exceeding the octet cannot forbid their existence. Nitrogen in NH_3 holds eight valence electrons (three bonding pairs plus a lone pair), not ten, and phosphorus in PCl_5 has ten electrons around it, not an octet's eight.

41. Which of the following statements about the halogens (Group 17) is correct?

- A) Bromine displaces chlorine from potassium chloride solution, because bromine is the stronger oxidising agent.
- B) Their boiling points decrease down the group, so astatine is a gas.
- C) Their reactivity increases down the group because the outer shell is further from the nucleus.
- D) They are all good electrical conductors in the liquid state.
- E) Their reactivity decreases down the group because the atoms get larger and attract an additional electron less strongly.

Answer: E

Halogens react as oxidising agents by gaining one electron; descending the group, the atomic radius and inner shell shielding increase, so the attraction for an added electron weakens and reactivity falls from fluorine to iodine. Consequently the more reactive halogen always displaces the less reactive one from its salts: chlorine displaces bromine from bromide solutions, and bromine cannot displace chlorine from potassium chloride. Boiling points rise down the group (chlorine is a gas, bromine a liquid, iodine a solid), and halogens are poor electrical conductors.

42. The standard enthalpies of formation of methane $\text{CH}_4(\text{g})$, carbon dioxide $\text{CO}_2(\text{g})$ and liquid water $\text{H}_2\text{O}(\text{l})$ are -75, -394 and -286 kJ/mol respectively. What is the enthalpy change for the complete combustion of one mole of methane: $\text{CH}_4(\text{g}) + 2\text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l})$? (The enthalpy of formation of $\text{O}_2(\text{g})$ is zero.)

- A) -1041 kJ/mol B) -891 kJ/mol C) -605 kJ/mol D) +605 kJ/mol E) +891 kJ/mol

Answer: B

$$\Delta H = \sum \Delta H_f(\text{products}) - \sum \Delta H_f(\text{reactants}) = [(-394) + 2 \times (-286)] - [(-75) + 0] = (-394 - 572) + 75 = -966 + 75 = -891 \text{ kJ/mol}$$

Dropping the factor of 2 on the water term gives $-394 - 286 + 75 = -605$, and subtracting the methane term with the wrong sign gives $-394 - 572 - 75 = -1041$; both are bookkeeping slips on the same equation. The positive values cannot be right, since a combustion is always strongly exothermic.

43. Hydrogen and iodine reach equilibrium: $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2\text{HI}(\text{g})$. At equilibrium the concentrations are $[\text{H}_2] = 0.20 \text{ mol/dm}^3$, $[\text{I}_2] = 0.30 \text{ mol/dm}^3$ and $[\text{HI}] = 0.60 \text{ mol/dm}^3$. What is the value of K_c ?

- A)** 0.17 **B)** 1.2 **C)** 2.45 **D)** 6.0 **E)** 10

Answer: D

For $\text{H}_2 + \text{I}_2 \rightleftharpoons 2\text{HI}$, $K_c = [\text{HI}]^2 / ([\text{H}_2] \times [\text{I}_2]) = (0.60)^2 / (0.20 \times 0.30) = 0.36 / 0.060 = 6.0$

The coefficient 2 in front of HI means its concentration enters squared; using [HI] unsquared gives $0.60/0.060 = 10$, and inverting the whole fraction gives about 0.17.

44. For the reaction $\text{A} + \text{B} \rightarrow \text{products}$ at 25°C , the following initial rates were measured. Experiment 1: $[\text{A}] = 0.10 \text{ mol/dm}^3$, $[\text{B}] = 0.10 \text{ mol/dm}^3$, rate = $0.0020 \text{ mol/(dm}^3 \text{ s)}$. Experiment 2: $[\text{A}] = 0.20 \text{ mol/dm}^3$, $[\text{B}] = 0.10 \text{ mol/dm}^3$, rate = $0.0040 \text{ mol/(dm}^3 \text{ s)}$. Experiment 3: $[\text{A}] = 0.10 \text{ mol/dm}^3$, $[\text{B}] = 0.20 \text{ mol/dm}^3$, rate = $0.0080 \text{ mol/(dm}^3 \text{ s)}$. What is the rate law for this reaction?

- A)** rate = $k[\text{A}][\text{B}]^2$
B) rate = $k[\text{A}]^2[\text{B}]^2$
C) rate = $k[\text{B}]^2$
D) rate = $k[\text{A}][\text{B}]$
E) rate = $k[\text{A}]^2[\text{B}]$

Answer: A

Step 1: with [B] held at 0.10, doubling [A] from 0.10 to 0.20 doubles the rate from 0.0020 to 0.0040, so the reaction is first order in A. Step 2: with [A] held at 0.10, doubling [B] from 0.10 to 0.20 quadruples the rate from 0.0020 to 0.0080, so the reaction is second order in B. Step 3: rate = $k[\text{A}][\text{B}]^2$, overall third order. Any law with B only to the first power would predict just a doubling of rate between the first and third experiments, not the observed quadrupling.

45. An alkane consists of a chain of five carbon atoms, with one methyl group attached to the second carbon of the chain and another methyl group attached to the fourth carbon. What is the IUPAC name of this alkane?

- A)** 2-methylhexane
B) 2,3-dimethylpentane
C) 2,4-dimethylpentane
D) 2,4-dimethylhexane
E) 3,5-dimethylpentane

Answer: C

The longest continuous carbon chain contains five atoms, so the parent alkane is pentane; no path through six carbons exists, so every hexane based name is impossible. The two methyl branches sit on carbons 2 and 4, and the structure is symmetric, so this numbering holds from either end of the chain, giving 2,4-dimethylpentane. A name placing the substituents at positions 3 and 5 is never acceptable, because the chain must always be numbered to give the lowest possible locants.

46. Methane reacts with chlorine gas in sunlight: $\text{CH}_4 + \text{Cl}_2 \rightarrow \text{CH}_3\text{Cl} + \text{HCl}$, followed by further substitution steps. What type of reaction is the first of these steps, $\text{CH}_4 \rightarrow \text{CH}_3\text{Cl}$?

- A)** an electrophilic addition
- B)** an elimination
- C)** a nucleophilic substitution
- D)** a condensation
- E)** a free-radical substitution

Answer: E

Sunlight (ultraviolet) splits Cl_2 into chlorine radicals; a chlorine radical abstracts a hydrogen atom from methane to give HCl and a methyl radical, which then reacts with another Cl_2 molecule to form CH_3Cl and regenerate a chlorine radical. One hydrogen of methane is replaced by chlorine with the carbon skeleton unchanged, which is the definition of free radical substitution. It is not electrophilic addition (methane has no double bond to add across), not nucleophilic substitution (no nucleophile attacks a carbon centre), and not an elimination.

47. What is the electron configuration of an iron(II) ion, Fe^{2+} ? (Iron has atomic number 26.)

- A)** $1s^2 2s^2 2p^6 3s^2 3p^6 3d^6$
- B)** $1s^2 2s^2 2p^6 3s^2 3p^6 3d^4 4s^2$
- C)** $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5 4s^1$
- D)** $1s^2 2s^2 2p^6 3s^2 3p^6 3d^8$
- E)** $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^6$

Answer: A

Neutral iron ($Z = 26$) fills as $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^6$

Fe^{2+} forms by losing the two electrons of highest energy, which are the 4s electrons, leaving $1s^2 2s^2 2p^6 3s^2 3p^6 3d^6$ with 24 electrons. Removing 3d electrons while keeping 4s reverses the order in which electrons are lost on ionisation, and the arrangement ending $4s^2 3d^6$ still belongs to the neutral atom, not the ion.

Section D - Mathematics (Q48-54)

48. Solve the equation $\log_2(x) + \log_2(x - 6) = 4$.

- A)** $x = 16$
- B)** There is no solution
- C)** $x = 8$ or $x = -2$
- D)** $x = -2$ only
- E)** $x = 8$ only

Answer: E

Step 1: combine the logarithms: $\log_2(x(x - 6)) = 4$, so $x(x - 6) = 2^4 = 16$

Step 2: rearrange to $x^2 - 6x - 16 = 0$, which factors as $(x - 8)(x + 2) = 0$, giving $x = 8$ or $x = -2$. Step 3: the logarithms require $x > 6$ (both x and $x - 6$ must be positive), so the negative root is rejected and $x = 8$ is the only solution; check: $\log_2 8 + \log_2 2 = 3 + 1 = 4$.

Keeping $x = -2$ ignores the domain of the logarithm

49. A committee of 3 people is chosen at random from a group of 4 chemists and 6 physicists. What is the probability that the committee contains exactly one chemist?

- A) $1/6$ B) $1/2$ C) $3/10$ D) $3/5$ E) $5/6$

Answer: B

Total committees = $C(10,3) = 120$

Favourable committees = 1 chemist chosen from 4 times 2 physicists chosen from 6 = $C(4,1) \times C(6,2) = 4 \times 15 = 60$.

Probability = $60/120 = 1/2$

Quoting the raw count of 60 without dividing by the 120 possible committees, or mixing ordered and unordered counts, produces the other fractions offered.

50. If $f(x) = x^2 - 4x$, which of the following expressions is equal to $f(x + 3) - 2$ for every value of x ?

- A) $x^2 - 4x + 1$ B) $x^2 - 4x - 5$ C) $x^2 + 2x - 1$ D) $x^2 + 2x - 5$ E) $x^2 - 10x + 19$

Answer: D

$f(x + 3) = (x + 3)^2 - 4(x + 3) = x^2 + 6x + 9 - 4x - 12 = x^2 + 2x - 3$, and subtracting 2 gives $x^2 + 2x - 5$

The expression $x^2 - 10x + 19$ is what substitution of $x - 3$ instead of $x + 3$ would produce, a sign slip in the horizontal shift.

51. How many values of x satisfy the equation $\cos(2x) = 1/2$ in the interval $0 \leq x \leq 2\pi$?

- A) 1 B) 2 C) 3 D) 4 E) 6

Answer: D

Put $t = 2x$; as x runs from 0 to 2π , t runs from 0 to 4π , two complete circuits of the unit circle. In each circuit, $\cos t = 1/2$ at $t = \pi/3$ and $t = 5\pi/3$, giving $t = \pi/3, 5\pi/3, 7\pi/3, 11\pi/3$ and hence $x = \pi/6, 5\pi/6, 7\pi/6, 11\pi/6$: four solutions. Counting solutions only over one period of cosine gives two, forgetting that doubling the angle doubles the range that must be searched.

52. Solve the inequality $(x + 1)/(x - 2) \geq 0$.

- A) $-1 \leq x < 2$
B) $x \leq -1$ or $x \geq 2$
C) $x \geq -1$ and $x < 2$
D) $x < -1$ or $x > 2$
E) $x \leq -1$ or $x > 2$

Answer: E

Step 1: the fraction equals 0 at $x = -1$, which is allowed since 0 satisfies ≥ 0 , and is undefined at $x = 2$, which is excluded. Step 2: sign test: for $x < -1$ numerator and denominator are both negative, so the fraction is positive; for $-1 < x < 2$ the signs differ, so it is negative; for $x > 2$ both are positive. Step 3: the solution set is $x \leq -1$ or $x > 2$. Writing $x \geq 2$ wrongly includes $x = 2$, where the fraction has no value.

53. A square is inscribed in a circle of radius 6 cm. What is the area of the square?

- A)** 24 cm² **B)** 36 cm² **C)** 72 cm² **D)** 108 cm² **E)** 144 cm²

Answer: C

A square inscribed in a circle has each diagonal equal to the circle's diameter: $d = 2 \times 6 = 12$ cm.

$$\text{Area} = d^2/2 = 12^2/2 = 144/2 = 72 \text{ cm}^2$$

Taking 6 cm as the side length gives 36 cm², but a square of side 6 has a diagonal of about 8.5 cm and would sit well inside a circle of radius 6 instead of touching it.

54. In an arithmetic sequence the third term is 11 and the eighth term is 31. What is the sum of the first ten terms?

- A)** 105 **B)** 210 **C)** 290 **D)** 390 **E)** 420

Answer: B

Step 1: from the two given terms, the common difference is $d = (31 - 11)/(8 - 3) = 20/5 = 4$.

Step 2: the first term is $a = 11 - 2 \times 4 = 3$

Step 3: $S_{10} = (10/2)(2a + 9d) = 5 \times (6 + 36) = 5 \times 42 = 210$

Equivalently the ten terms run 3, 7, ..., 39 with mean $(3 + 39)/2 = 21$, and $21 \times 10 = 210$.

Section E - Physics (Q55-60)

55. A ball is kicked from level ground with an initial speed of 20 m/s at an angle of 30 degrees above the horizontal. What is the maximum height it reaches? ($g = 10 \text{ m/s}^2$; $\sin 30 \text{ degrees} = 0.5$; $\cos 30 \text{ degrees} = 0.87$; ignore air resistance.)

- A)** 2.5 m **B)** 5 m **C)** 10 m **D)** 20 m **E)** 40 m

Answer: B

The vertical launch component is $v_{0y} = 20 \times \sin 30^\circ = 20 \times 0.5 = 10 \text{ m/s}$

At maximum height the vertical velocity is momentarily zero, so from $v^2 = u^2 - 2gh$: $h = 10^2/(2 \times 10) = 100/20 = 5 \text{ m}$. Squaring the full launch speed of 20 m/s instead of its vertical component gives 20 m, the classic error behind that distractor.

56. A ball of mass 4.0 kg rolls along level ground with kinetic energy 200 J. What is its speed?

- A)** 5 m/s **B)** 10 m/s **C)** 25 m/s **D)** 50 m/s **E)** 100 m/s

Answer: B

Kinetic energy $= \frac{1}{2}mv^2$ gives $200 = \frac{1}{2} \times 4.0 \times v^2 = 2v^2$, so $v^2 = 100$ and $v = 10 \text{ m/s}$

The other speeds fail the same relation: 5 m/s would give only $\frac{1}{2} \times 4.0 \times 25 = 50 \text{ J}$, and 25 m/s would give 1250 J.

57. On a frictionless horizontal track a trolley of mass 2.0 kg moving at 6.0 m/s collides head-on and elastically with a stationary trolley of mass 4.0 kg. What is the speed of the 4.0 kg trolley immediately after the collision?

- A)** 2 m/s **B)** 3 m/s **C)** 4 m/s **D)** 6 m/s **E)** 12 m/s

Answer: C

Step 1: momentum conservation: $2.0 \times 6.0 = 2.0v_1 + 4.0v_2$

Step 2: in a head on elastic collision the relative speed of separation equals the relative speed of approach: $v_2 - v_1 = 6.0$. Step 3: solving the pair gives $v_1 = -2$ m/s (the light trolley rebounds) and $v_2 = 4.0$ m/s.

Check energy: before, $\frac{1}{2} \times 2 \times 6^2 = 36$ J; after, $\frac{1}{2} \times 2 \times 2^2 + \frac{1}{2} \times 4 \times 4^2 = 4 + 32 = 36$ J

A speed of 6 m/s for the heavy trolley would more than double the system's kinetic energy, violating energy conservation.

58. An object of weight 50 N and density 2500 kg/m³ hangs fully submerged in water of density 1000 kg/m³, held at rest by a vertical string. What is the tension in the string? ($g = 10$ m/s².)

- A)** 12.5 N **B)** 20 N **C)** 30 N **D)** 50 N **E)** 70 N

Answer: C

The object's mass is $50/10 = 5$ kg and its volume is $5/2500 = 0.002$ m³, so the upthrust equals the weight of displaced water: $1000 \times 10 \times 0.002 = 20$ N. Held at rest, the string tension balances weight minus upthrust: $T = 50 - 20 = 30$ N. Ignoring buoyancy gives 50 N, and adding the upthrust instead of subtracting it gives 70 N, the wrong sign for a force that pushes upward.

59. How much heat is needed to raise the temperature of 0.30 kg of olive oil from 20 degrees C to 70 degrees C? The specific heat capacity of the oil is 2000 J/(kg K); ignore heat losses.

- A)** 3.0 kJ **B)** 15 kJ **C)** 30 kJ **D)** 60 kJ **E)** 120 kJ

Answer: C

The temperature rise is $\Delta T = 70 - 20 = 50$ K

$Q = mc\Delta T = 0.30 \times 2000 \times 50 = 30\,000$ J = 30 kJ

The value 3.0 kJ would follow from a tenfold slip in the mass or the temperature rise, and using an absolute temperature such as 20 K or 70 K instead of the rise also misses the mark.

60. A 12 V battery of negligible internal resistance is connected directly across a single 8 Ω resistor. What is the power dissipated in the resistor?

- A)** 1.5 W **B)** 18 W **C)** 24 W **D)** 96 W **E)** 144 W

Answer: B

$P = V^2/R = 12^2/8 = 144/8 = 18$ W

Equivalently, $I = V/R = 12/8 = 1.5$ A and $P = VI = 12 \times 1.5 = 18$ W

The number 1.5 is the current in amperes rather than a power, 96 W comes from multiplying V by R, and 144 W from stopping at V^2 without dividing by the resistance.